Chapter - 2

Ahmdahl's law

little law:

↳ queuing process

system in → steady / equilibrium → continuously giving service at the same rate.



(Servers / transmission line)

Wanna know how many packages pending.

items in the system while its giving service continuously.

↳ $L = \lambda * \omega$

λ = Average arrival rate of item

ω = " Service time (staying)

Example: In a book store, 10 visitor arrive every hour → λ

30 min by visitor → ω

$$L = \lambda \times \omega = 10 \times 30 \text{ min}$$

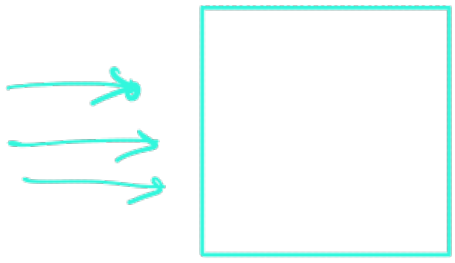
$$= 10 \times \frac{1}{2} \text{ hr} = 5 \text{ visitor/hr.}$$

Should you modify the business plan?

Clock speed:

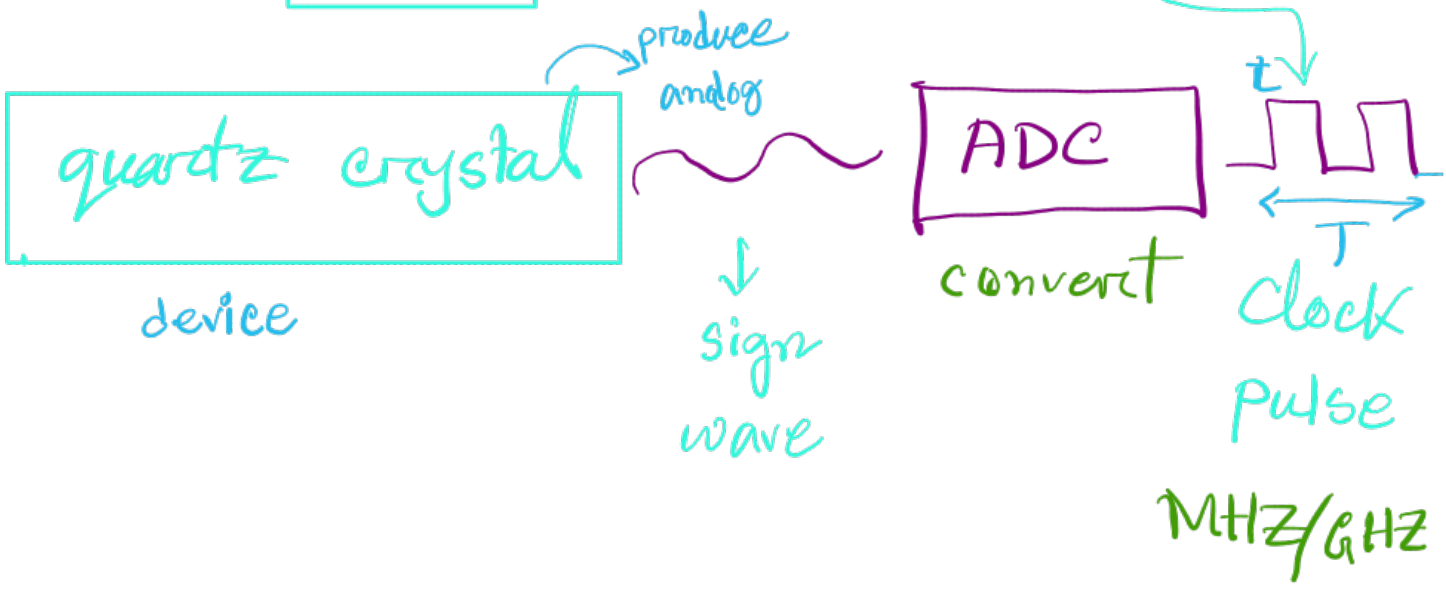
processor's heartbeat.

mem r/w, instr execution synchronized with this clock.



Duty cycle = ——

= t/T



time for synchronization; Time taken to settle down to 0/1 value.

Number of cycles for instr. execution

↳ I/O → most 20/30 ↳ to settle down

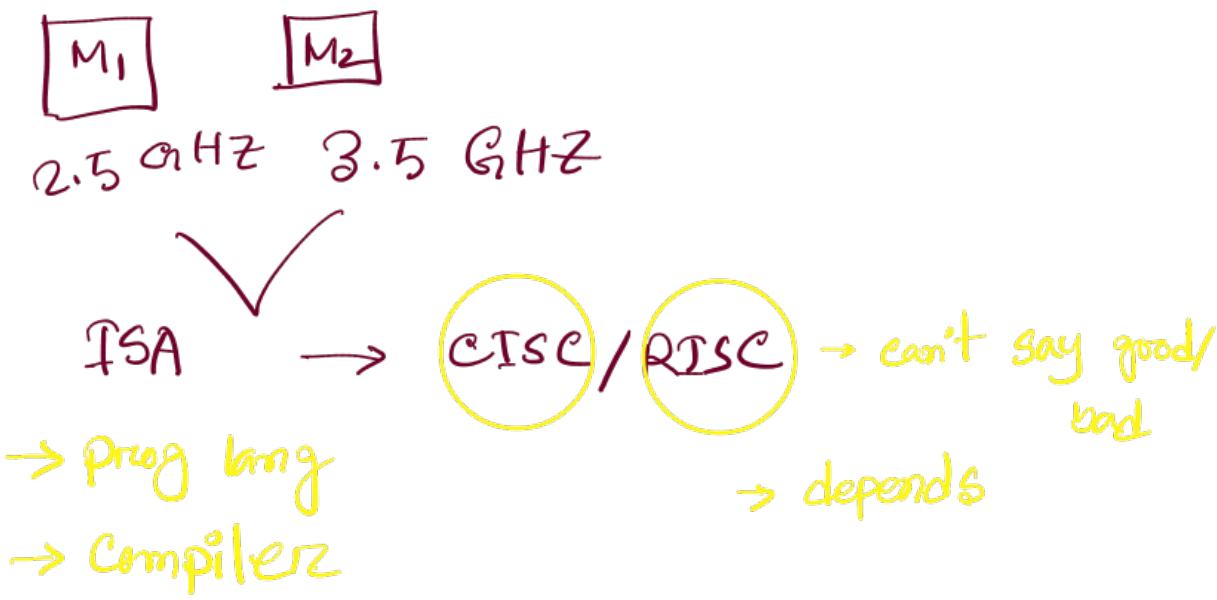
to a particular signal

mem → 0 to 10 CP.

T = cycle time = time to execute a cycle.

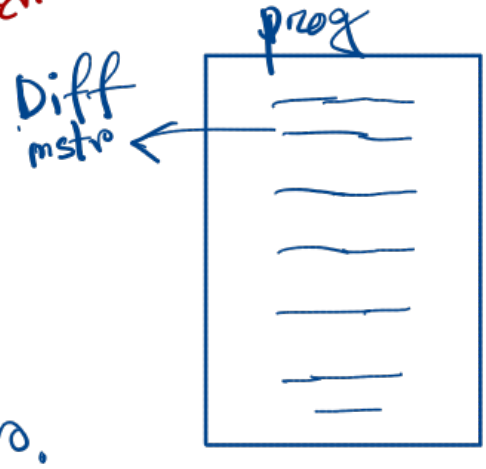
Duty cycle = $\frac{\text{how much time clock in high state}}{\text{total time}}$

clock pulse not enough to determine.



Another Matric → Instro. execution rate. → T^{-1}

At a time, how many n executed.



τ = cycle time

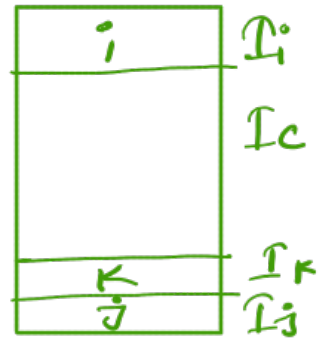
freq. $f = 1/\tau$

CPI = cycles per instro.

instr → related to processor → fixed cycles required

x cycles = Processor add sub
y " = mem r/w
z = I/O

x < y < z



Total
↳ Ic number of instr-

Diff instr → Diff rate

$$I_c = \sum_{i=1}^{n} I_i$$

↳ n type processor.

Ic = (i) type of instr.
↳ (i) → CPIi

$$CPI = \frac{\sum_{i=1}^{n} I_i \times CPI_i}{I_c}$$ → numerator / C

$$T = \left(\sum I_i \times CPI_i\right) \times C = I_c \times CPI \times C$$

↓ Time to complete Ic no of instr.

↳ Avg CPI calculate.

I

2 parts → mem | proc

m

mem cycles

$$T = I_c \times \left(p + m * k \right) * \tau$$

p: very small

m: very big

k: normalization const (adjust p & m.)

p: (cycles for processor)

m: (cycle alloted for memory)

Ins^n per second (IPS):

$$\text{MIPS rate} = \frac{I_c}{T \times 10^6} = \frac{1}{CPI \times \tau \times 10^6}$$

(IPS)

$$= \frac{I_c}{I_c \times CPI \times \tau \times 10^6}$$

$$= \frac{1}{CPI \times \tau \times 10^6} = \frac{f}{CPI \times 10^6}$$

A number to represent a single system.

mean deviations → rep of [sys]

Mean → pythogorean

→ geometric

arithmatic Harmonic

$$AM = \frac{x_1 + x_2 + \cdots x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

independent & equally imp.

If related in reciprocal way

so AM will ↑

$(x_1)^{+} \propto AM^{+}$

proportional.

$(x_1)^{+} \propto \frac{1}{(Mean)^{-}}$

reciprocal/Harmonic

$$HM = \left(\frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots \frac{1}{x_n}} \right) = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

$\frac{x_1 + x_2 \cdots}{n}$. reciprocal → $\frac{n}{\frac{1}{x_1}}$

↓

velocity, capacitance in series, Resist. in parallel.

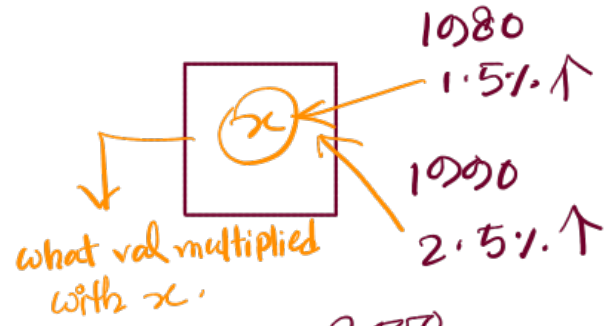
$\overline{V} = \frac{V_1 + V_2 + V_3}{}$ ✗

$$= \frac{3}{1/v_1 + 1/v_2 + 1/v_3}$$

Harmonic →

$v = \frac{s}{t}$ (fixed) → AM wrong
(diff) → HM correct

GM: Compounded growth.



to calc avg growth of population → always increasing.

indep.

$$\frac{x_1 + x_2 + x_3 \cdots}{n}$$

recip

$$\frac{n}{\frac{1}{x_1} + \frac{1}{x_2} \cdots}$$

↓

Chained Operation

gotta consider prev value always

$$\text{So, GM} = \sqrt[n]{x_1 \cdot x_2 \cdots x_n}$$

$$= \sqrt[4]{1.018 \times 1.025 \times 1.035 \times 1.08}$$

$$= x + 1.5\%$$

$x \rightarrow$ 1.5% (initial)

growth = $x * 1.015$

Total = $x + x * 1.015$

avg growth per decades.

Relation of the 3 M.→ FM

$$FM = f^{-1}\left(\frac{f(x_1) + f(x_2) + \cdots\cdots f(x_n)}{n}\right)$$

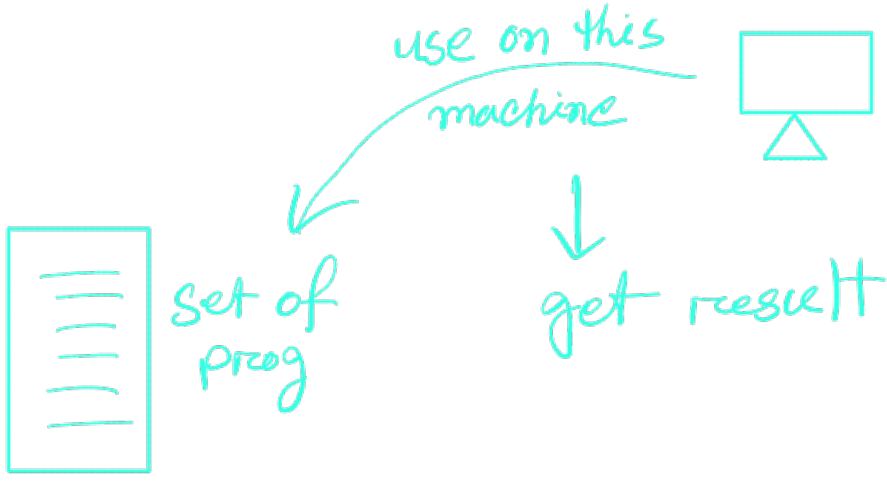
$f(x) = x \longrightarrow AM$

$f(x) = 1/x \rightarrow HM$

$f(x) = \ln x \rightarrow GM$

$HM \leq GM \leq AM$

$GM^2 = AM * HM$



↳ Bench Mark → 1080s

Commercial operation.

If diff company uses diff diff bench mark can't be compared.

Standard → SPEC : cloud → 2018

CPU → 2017

Storage → 2020

power consumption - 2008

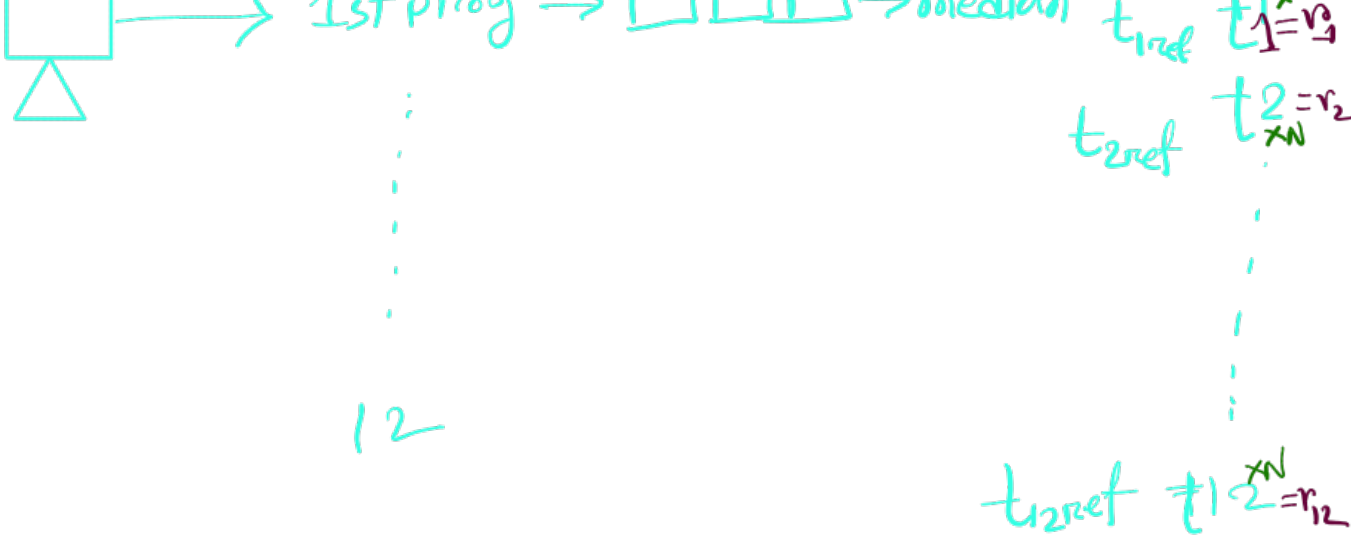
2021

spec 2006

12 int. prog	17 float type prog.

check ref machine value → speed

|| GM → best for comparison.

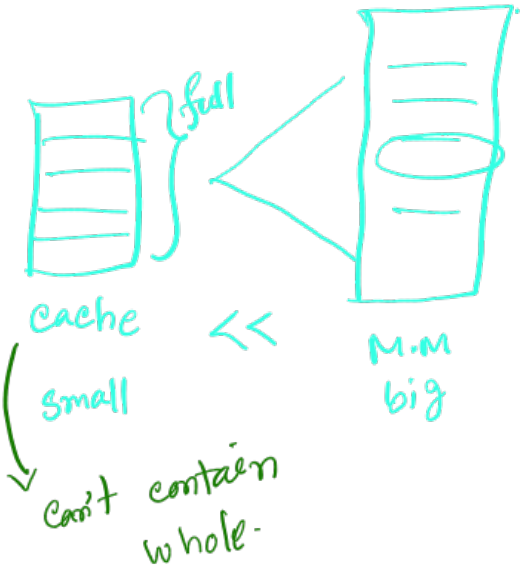


$$\text{mean} = \sqrt[12]{v_1 \cdot v_2 \cdots v_{12}}$$

$$\text{for float} = \sqrt[17]{v_1 \cdot v_2 \cdots v_{17}}$$

$\Rightarrow$ N Processors.

Chapter 4 Replacement Algorithm



So. 3 kinds of Mapping

$j = i \% m$ → D.M (Direct Mapping)
→ FAM
→ SAM

LRU - Least Recently used. (linear)

LFU Least frequently used

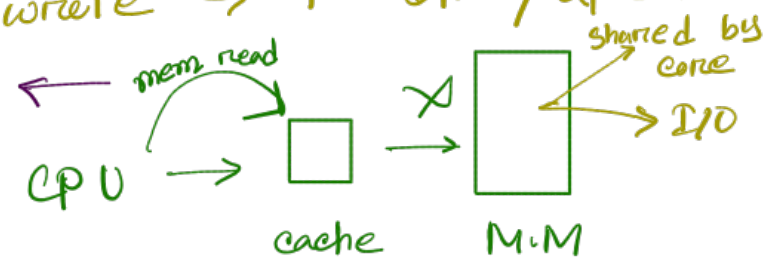
Replace FIFO

Similar → Random

Write policy : Cache write

memory write → to store/update data.

if found (cache hit) ← mem read



write → cache 1st.

to confirm integrity → 2 operations.

1. write through → change a single bit
2. " back

→ Huge data transmission → Update

↳ Dirty bit check → 1

↳ Mem block updat

Cheap cache.

To maintain integrity :

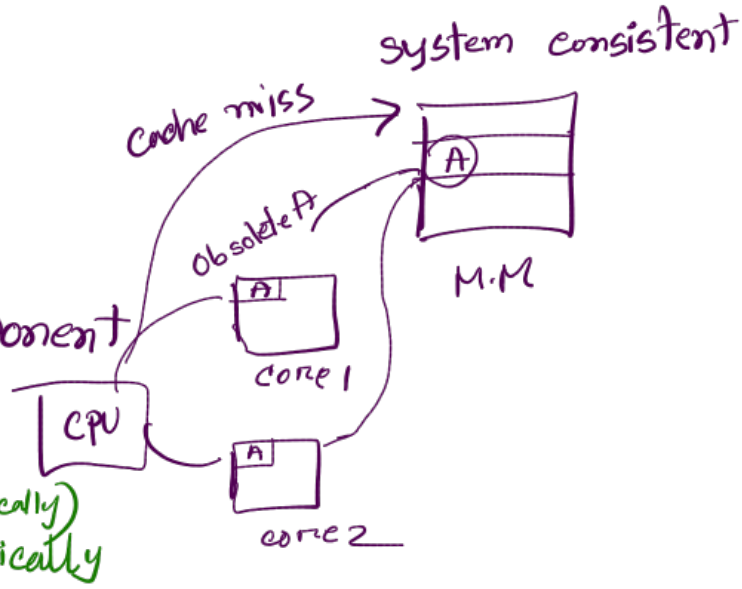
→ Bus watching approach

→ Hardware transparent component

→ non cacheble memory.

↳ line size ↑ (better theoritically) worse practically

→



line size ↑ → cache miss → small wouldve been easy

As it would take less

time to check the data?

If data inside cache → less time

" " in M.M → Increase Memory latency.